



ScholarNest

Spark Azure Databricks

Databricks Spark Certification and beyond

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Azure
Databricks

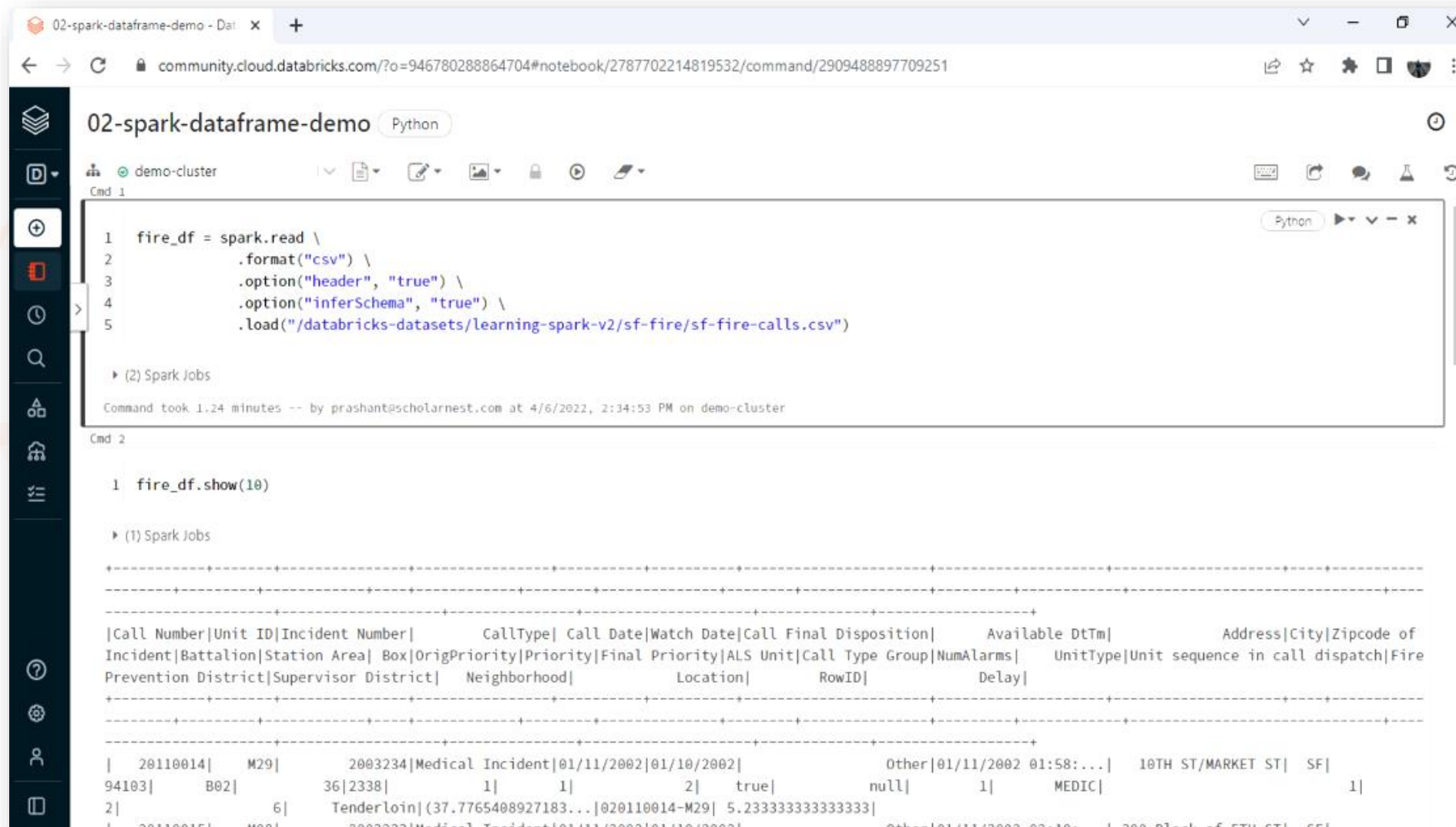


Absolute Beginner to Specialization in Apache Spark and Azure Databricks



Dataframe Methods

Go to your Databricks workspace and open spark-dataframe-demo notebook. **(Reference: 02-spark-dataframe-demo.ipynb)**



The screenshot shows a Databricks notebook interface with the title "02-spark-dataframe-demo" and the language set to "Python". The notebook is running on a cluster named "demo-cluster".

Cmd 1:

```
1 fire_df = spark.read \  
2     .format("csv") \  
3     .option("header", "true") \  
4     .option("inferSchema", "true") \  
5     .load("/databricks-datasets/learning-spark-v2/sf-fire/sf-fire-calls.csv")
```

(2) Spark Jobs

Command took 1.24 minutes -- by prashant@scholarnest.com at 4/6/2022, 2:34:53 PM on demo-cluster

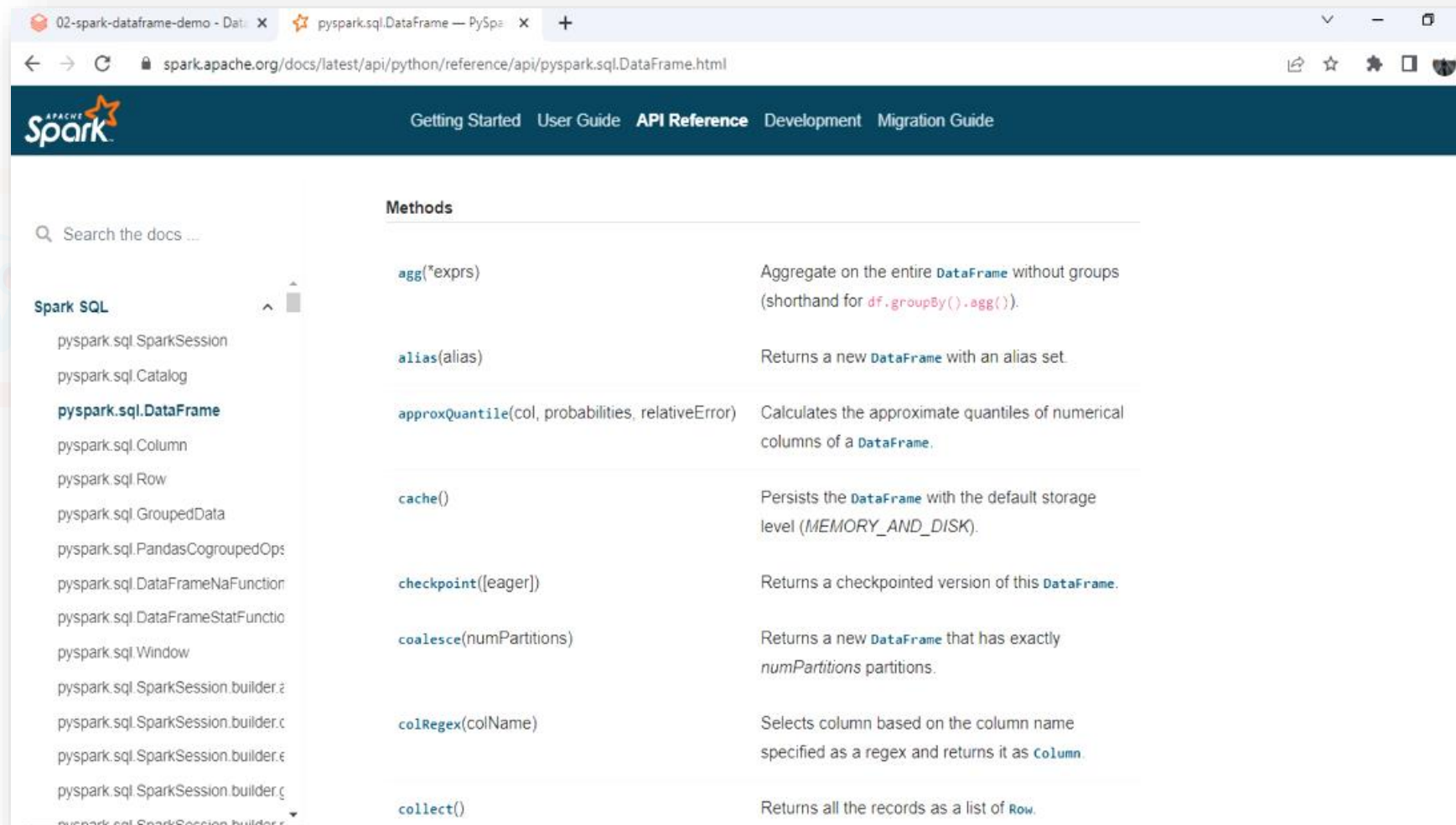
Cmd 2:

```
1 fire_df.show(10)
```

(1) Spark Jobs

Call Number	Unit ID	Incident Number	CallType	Call Date	Watch Date	Call Final Disposition	Available DtTm	Address	City	Zipcode of Incident
20110014	M29	2003234	Medical Incident	01/11/2002	01/10/2002	Other	01/11/2002 01:58:...	10TH ST/MARKET ST	SF	
94103	B02	36 2338	1	1	2	true	null	1	MEDIC	1
2		6	Tenderloin	37.7765408927183...	020110014-M29	5.233333333333333				
20110015	M00	2003232	Medical Incident	01/11/2002	01/10/2002	Other	01/11/2002 02:10:...	1 200 Block of ETH ST	SF	

Now that we have created a Dataframe, we can apply Dataframe methods to process the data. You can visit <https://spark.apache.org/docs/latest/api/python/reference/api/pyspark.sql.DataFrame.html> to get the list of all the Dataframe methods.



Dataframe offers you a long list of methods. However, we can classify them into three logical groups:

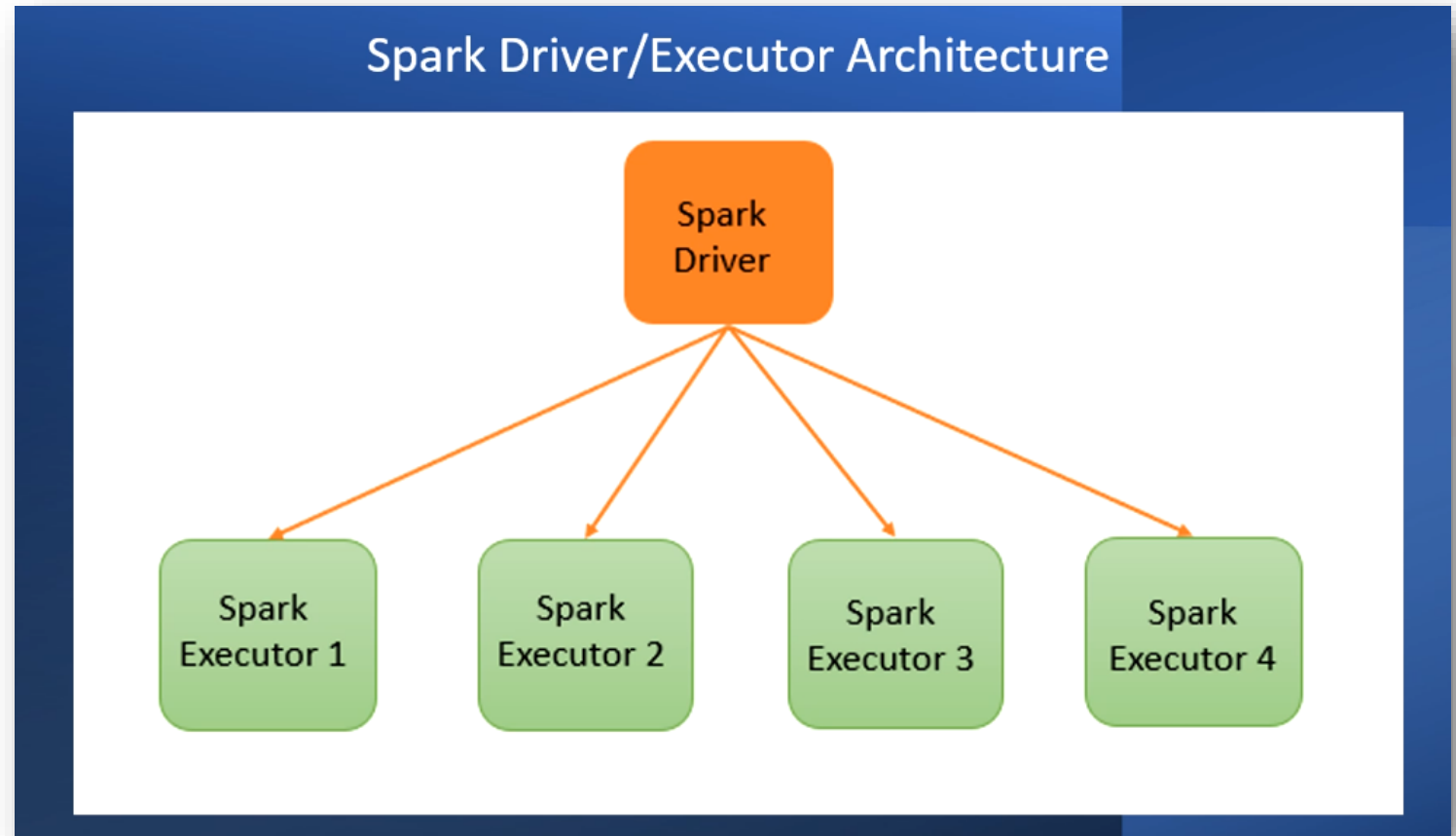
1. Actions
2. Transformations
3. Functions/Methods

Actions: Actions trigger a Spark Job and return to the Spark driver.

Spark is a distributed processing system. So every Spark program runs as one driver and one or more executors

And Actions are Dataframe operations that performs two things:

1. Kick off a Spark Job
2. Return to the Driver



Transformations: Spark Dataframe transformation produces a newly transformed Dataframe.

Transformations are different than actions. They do not kick off a Spark Job, and they do not return to the driver.

They simply create a new Dataframe and return it.

Functions and Methods : These are DataFrame methods or functions which are not categorized into Actions or Transformations.
So everything other than the Actions and Transformations is a method or function.

Here is a list of Dataframe Actions and Dataframe Transformations.
Spark offers you 12 actions as on date. This list may update with a newer version of Spark.
We have 38 transformations.

Actions

1. collect
2. count
3. describe
4. first
5. foreach
6. foreachPartition
7. head
8. show
9. summary
10. tail
11. take
12. toLocalIterator

Transformations

- | | | |
|---------------------|--------------------------|-----------------------|
| 1. agg | 16. intintersect | 31. subtract |
| 2. alias | 17. ersectAll | 32. transform |
| 3. coalesce | 18. join | 33. union |
| 4. colRegex | 19. limit | 34. unionAll |
| 5. crossJoin | 20. orderBy | 35. unionByName |
| 6. crosstab | 21. randomSplit | 36. where |
| 7. cube | 22. repartition | 37. withColumn |
| 8. distinct | 23. repartitionByRange | 38. withColumnRenamed |
| 9. drop | 24. rollup | |
| 10. drop_duplicates | 25. sample | |
| 11. dropDuplicates | 26. sampleBy | |
| 12. dropna | 27. select | |
| 13. exceptAll | 28. selectExpr | |
| 14. filter | 29. sort | |
| 15. groupby | 30. sortWithinPartitions | |

Here is a list of Dataframe Functions and Methods.

We have 20 functions and methods. These are utility functions or methods.

Functions/Methods

1. approxQuantile
2. cache
3. checkpoint
4. createGlobalTempView
5. createOrReplaceGlobalTempView
6. createOrReplaceTempView
7. createTempView
8. explain
9. hint
10. inputFiles
11. isLocal
12. localCheckpoint
13. persist
14. printSchema
15. registerTempTable
16. toDF
17. toJSON
18. unpersist
19. writeTo
20. withWatermark

There are ten more functions that I excluded from the list that I have highlighted in the image below. I have excluded these because those functions are only available in Python and not in other languages, and it is not frequently used.

Functions/Methods

- | | | |
|----------------------------------|-------------------|-----------------------|
| 1. approxQuantile | 16. toDF | 1. corr |
| 2. cache | 17. toJSON | 2. cov |
| 3. checkpoint | 18. unpersist | 3. freqItems |
| 4. createGlobalTempView | 19. writeTo | 4. mapInPandas |
| 5. createOrReplaceGlobalTempView | 20. withWatermark | 5. replace |
| 6. createOrReplaceTempView | | 6. sameSemantics |
| 7. createTempView | | 7. semanticHash |
| 8. explain | | 8. to_koalas |
| 9. hint | | 9. to_pandas_on_spark |
| 10. inputFiles | | 10. toPandas |
| 11. isLocal | | |
| 12. localCheckpoint | | |
| 13. persist | | |
| 14. printSchema | | |
| 15. registerTempTable | | |

Finally, here is the definition of all the three types of Dataframe Methods.

Dataframe Methods

1. Actions

Actions are Dataframe operations that **kick off a Spark Job** execution and **return to the Spark driver**

2. Transformations

Spark Dataframe transformation **produces a newly transformed Dataframe**

3. Functions/Methods

Dataframe methods or functions which are not categorized into Actions or Transformations



Thank You
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